STRENGTH OF MATERIALS

Subject Code : 23A01302T

UNIT – V

Columns, Thin and Thick Cylinders

Handwritten Study Notes

Syllabus topics covered in this unit :

Classification of columns — Euler's crippling load theory — end conditions — Equivalent length — Slenderness ratio — Limitations — Rankine-Gordon — Secant formula — Perry's formula — Thin & thick cylindrical shells — Hoop, longitudinal & volumetric strains — Lamé's theory — Compound cylinders

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→ 5.1 Columns – Introduction
→ 5.2 Euler's Crippling (Buckling) Load
→ 5.3 Rankine-Gordon Formula (Empirical)
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→ 5.6 Thick Cylinders – Lamé's Theory
→ 5.7 Compound Cylinders

5.1 Columns – Introduction

A column (strut) is a vertical compression member whose length is large compared with its lateral dimensions ; it fails by buckling (sudden lateral bending) before crushing.

Classification

- Short column : fails by crushing ; $\lambda < \sim 30$; $P = \sigma_c A$.
- Medium column : crushing + buckling ; Rankine/Johnson formulae.
- Long column : fails by buckling ; Euler's formula valid.
- Axially loaded (ideal) column vs eccentrically loaded column.

Slenderness ratio

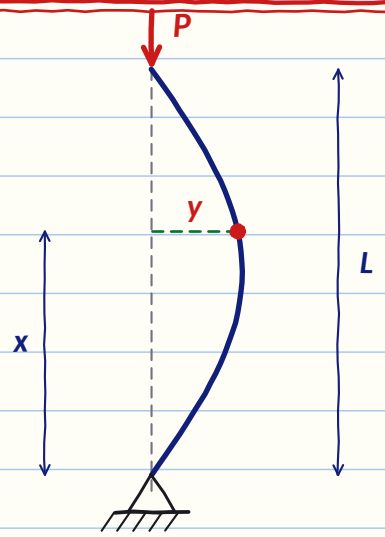
$$\lambda = \frac{L_e}{k} \qquad k = \sqrt{\left(\frac{I_{min}}{A}\right)} \quad \text{(least radius of gyration)}$$

- Column buckles about the axis of least I. L_e = equivalent (effective) length = length of the equivalent pin-ended column.

Assumptions of Euler's theory

- Column is perfectly straight, homogeneous, load strictly axial ; ends as specified.
- Material obeys Hooke's law ; stress ≤ proportional limit ; self-weight neglected ; failure only by buckling.
- Small deflections ; section uniform.

5.2 Euler's Crippling (Buckling) Load



At section x : $M = -P y$

$E I \, d^2y/dx^2 = -P y$

$d^2y/dx^2 + (P/EI)\, y = 0$

$y = A \cos \alpha x + B \sin \alpha x$, $\alpha^2 = P/EI$

$y(0) = 0 \Rightarrow A = 0$; $y(L) = 0 \Rightarrow \sin \alpha L = 0$

$\alpha L = n\pi \Rightarrow$ **$P_{cr} = \pi^2 E I / L^2$** $(n = 1)$

Derivation – both ends hinged

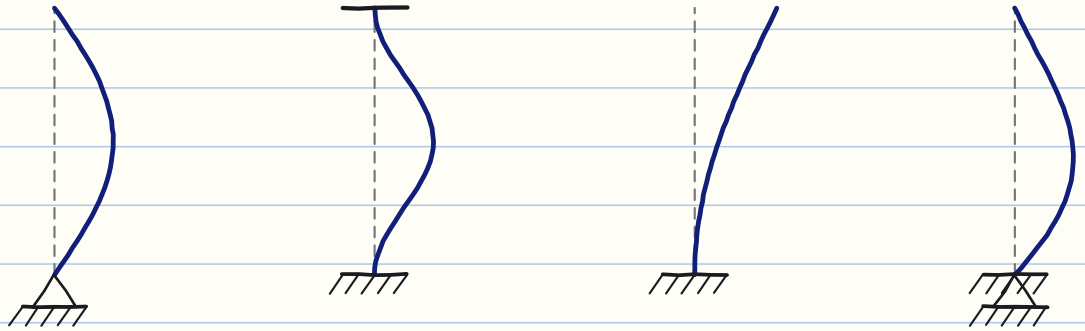
- Let y = deflection at distance x from the lower end ; moment $M = -P y$.
- $E I \, d^2y/dx^2 = -P y \Rightarrow d^2y/dx^2 + \alpha^2 y = 0$ with $\alpha^2 = P/EI$.
- Solution $y = A \cos \alpha x + B \sin \alpha x$. $y(0) = 0 \Rightarrow A = 0$; $y(L) = 0 \Rightarrow B \sin \alpha L = 0 \Rightarrow \alpha L$

$= n\pi$.

- Smallest load ($n = 1$) is the crippling load :

$$P_{cr} = \frac{\pi^2 E I}{L^2}$$

Other end conditions



Le = L	Le = L/2	Le = 2L	Le = L/√2
Both ends hinged	Both ends fixed	One fixed, one free	One fixed, one hinged

End conditions	L_e	Euler load P_{cr}
Both ends hinged	L	π^2EI / L^2
Both ends fixed	$L / 2$	$4\pi^2EI / L^2$
One fixed, other free	$2L$	$\pi^2EI / 4L^2$
One fixed, other hinged	$L/\sqrt{2}$	$2\pi^2EI / L^2$

$$P_{cr} = \frac{\pi^2 E I}{L_e^2}$$

- Fixed–free (derivation) : $EIy'' = P(\delta - y) \Rightarrow y = \delta(1 - \cos \alpha x)$; $y(L) = \delta \Rightarrow \cos \alpha L = 0 \Rightarrow \alpha L = \pi/2 \Rightarrow P = \pi^2EI/4L^2$.
- Ratio of strengths (hinged : fixed-fixed : fixed-free : fixed-hinged) = 1 : 4 : ¼ : 2.

Euler's critical stress

$$\sigma_{cr} = \frac{P_{cr}}{A} = \frac{\pi^2 E}{\lambda^2} \qquad \text{(hyperbola in } \sigma - \lambda \text{ plane)}$$

Limitations of Euler's formula

- Valid only while $\sigma_{cr} \le$ proportional limit $\Rightarrow \lambda \ge \lambda_c = \pi\sqrt{(E/\sigma_p)}$.
- For mild steel ($E = 2\times10^5$, $\sigma_p \approx 330$ N/mm^2) : $\lambda_c = \pi\sqrt{(2\times10^5/330)} = 77$; Euler applies for $\lambda > \sim 80$ (cast iron ~ 80, timber ~ 60).
- For small λ Euler gives absurdly high stress (> yield) ⇒ use Rankine/Johnson/Perry.
- Ignores initial crookedness, eccentricity, residual stress and imperfect end fixity.

Example 1 (Euler loads)

Solid circular column $d = 50$ mm, length 3 m, $E = 2\times10^5$ N/mm^2.

$I = \pi d^4/64 = 3.068\times10^5$ mm^4 ; $k = d/4 = 12.5 \Rightarrow \lambda = 3000/12.5 = 240$ (long ✓)

Both ends hinged : $P_{cr} = \pi^2 \times 2\times10^5 \times 3.068\times10^5 / 3000^2 =$ **67.3 kN**

Both ends fixed : $4 \times 67.3 =$ **269.2 kN** ; fixed–free : $67.3/4 =$ **16.8 kN** ;

fixed–hinged : $2 \times 67.3 =$ **134.6 kN**

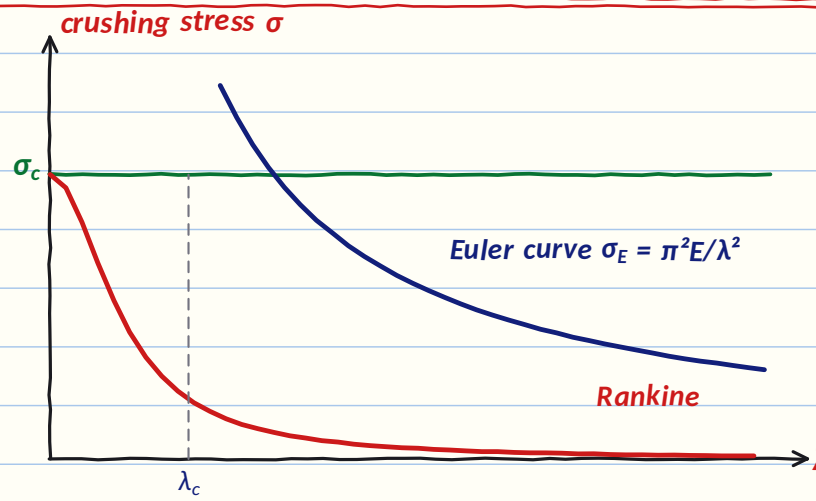
Example 2 (Hollow / rectangular section)

Rectangular column 60 × 100 mm, 2.5 m, both ends hinged, $E = 2\times10^5$:

$I_{min} = 100 \times 60^3/12 = 1.8\times10^6$ mm^4 (buckles about the weaker axis)

$P_{cr} = \pi^2 \times 2\times10^5 \times 1.8\times10^6 / 2500^2 =$ **568.4 kN**

5.3 Rankine–Gordon Formula (Empirical)



Rankine–Gordon :

$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_E}$

$P_R = \sigma_c A /(1 + a\,\lambda^2)$

$a = \sigma_c /\pi^2 E$ (Rankine const.)

Short: λ small → $P \approx$ crushing

Long: λ large → $P \approx$ Euler

- Covers short, medium and long columns. Rankine's crippling load P_R :

$$\frac{1}{P_R} = \frac{1}{P_c} + \frac{1}{P_E} \qquad P_c = \sigma_c A \ \text{(crushing)}, \ P_E = \text{Euler load}$$

$$P_R = \frac{\sigma_c A}{1 + a\,\lambda^2} \qquad a = \frac{\sigma_c}{\pi^2 E}$$

Material	σ_c (N/mm^2)	a
Mild steel	330	1/7500
Cast iron	550	1/1600
Wrought iron	250	1/9000
Timber	50	1/750

- λ small ⇒ $P_R \to \sigma_c A$ (crushing) ; λ large ⇒ $P_R \to$ Euler load.

Example 3 (Rankine)

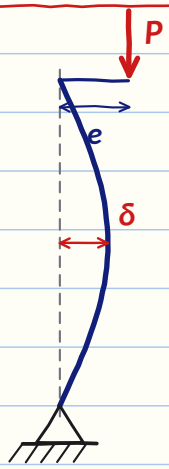
Solid circular steel column $d = 100$ mm, 3 m, both ends hinged, $\sigma_c = 330$ N/mm^2, $a = 1/7500$.

$A = 7854$ mm^2 ; $k = 25$ mm ; $\lambda = 3000/25 = 120$

$P_R = 330 \times 7854 / (1 + 120^2/7500) = 2.592\times10^6 / 2.92 =$ **887.6 kN**

Safe load (F.O.S. 3) = **295.9 kN**

5.4 Eccentric Loading – Secant Formula



Max deflection $\delta = e\,[\,\sec(\tfrac{1}{2}\sqrt{(P\,L^2/EI)}) - 1\,]$

Max moment $M_{max} = P\,(e + \delta)$

$$\sigma_{max} = \frac{P}{A}\left(1 + \frac{e\,c}{k^2}\sec\left(\frac{L}{2k}\sqrt{\frac{P}{AE}}\right)\right)$$

SECANT FORMULA (pin-ended, $c = y_{max}$)

- Load P acts at eccentricity e from the axis of a pin-ended column ; bending moment at mid-height = $P\,(e + \delta)$.

$$\delta = e\left[\sec\left(\frac{L}{2}\sqrt{\frac{P}{EI}}\right) - 1\right]$$

$$\sigma_{max} = \frac{P}{A}\left[1 + \frac{e\,c}{k^2}\sec\left(\frac{L_e}{2k}\sqrt{\frac{P}{A\,E}}\right)\right]$$

- c = distance of extreme fibre from the axis ; $e\,c/k^2$ = eccentricity ratio. Failure when $\sigma_{max} = \sigma_y$.
- Because P appears inside sec, the equation is solved by trial for P.

Example 4 (Secant)

Pin-ended column 100 × 100 mm, 3 m, $P = 100$ kN, $e = 10$ mm, $E = 2\times10^5$.

$k^2 = 100^2/12 = 833.3$; $c = 50$; $e\,c/k^2 = 0.6$

$(L/2k)\sqrt{(P/AE)} = (3000/57.74) \times \sqrt{(10^5/(10^4\times2\times10^5))} = 0.3674$ rad $\Rightarrow \sec = 1.0715$

$\sigma_{max} = 10\,[1 + 0.6 \times 1.0715] =$ **16.4 N/mm^2**

Perry's formula (Perry–Robertson)

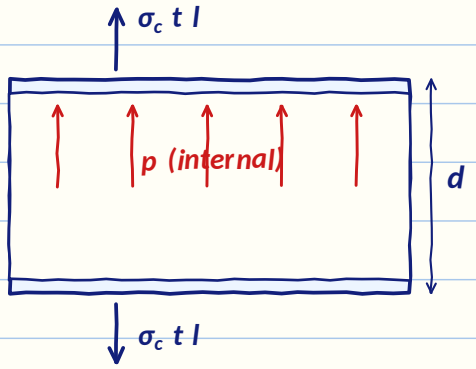
$$(\sigma_y - \sigma_a)(\sigma_E - \sigma_a) = \eta\, \sigma_E\, \sigma_a$$

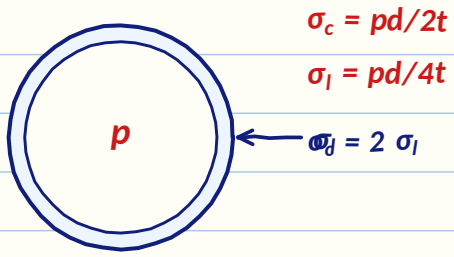
- σ_a = average axial stress at failure = P/A ; σ_y = yield stress ; σ_E = Euler stress = $\pi^2 E/\lambda^2$.
- η = Perry factor = $0.3\,(\lambda/100)^2$ (allows for initial curvature / eccentricity) – basis of many design codes.
- Solution : $\sigma_a = \tfrac{1}{2}[\,\sigma_y + (1+\eta)\,\sigma_E - \sqrt{((\sigma_y + (1+\eta)\sigma_E)^2 - 4\,\sigma_y\,\sigma_E)}\,]$.

5.5 Thin Cylindrical Shells

Thin shell : wall thickness t < d/20 ; stress assumed uniform across thickness.

Internal pressure p.





Half cylinder: $2\,\sigma_c\,t\,l = p\,d\,l$ End: $\sigma_l\,\pi\,d\,t = p\,\pi\,d^2/4$

Circumferential (hoop) stress

- Consider a length l ; force from pressure on half-cylinder = $p\,d\,l$; resisted by $2\,\sigma_c\,t\,l$.

$$\sigma_c = \frac{p\,d}{2\,t}$$

Longitudinal stress

- End force $p\,\pi\,d^2/4$ resisted by wall area $\pi\,d\,t$:

$$\sigma_l = \frac{p\,d}{4\,t} = \frac{\sigma_c}{2}$$

- Max shear stress (in plane) = $(\sigma_c - \sigma_l)/2 = p\,d/8t$.

Strains and volume change (μ = Poisson's ratio)

$$\varepsilon_c = \frac{p\,d}{4\,t\,E}(2-\mu) \qquad \varepsilon_l = \frac{p\,d}{4\,t\,E}(1-2\mu)$$

$$\varepsilon_v = 2\,\varepsilon_c + \varepsilon_l = \frac{p\,d}{4\,t\,E}(5-4\mu)$$

- Change in diameter $\delta d = \varepsilon_c\,d$; change in length $\delta L = \varepsilon_l\,L$; $\delta V = \varepsilon_v\,V$.
- Joint efficiency : $\sigma_c = p\,d\,/\,(2\,t\,\eta_l)$ (longitudinal joint) ; $\sigma_l = p\,d\,/\,(4\,t\,\eta_c)$ (circumferential joint).
- Thin spherical shell : $\sigma = p\,d/4t$; $\varepsilon = p\,d/4tE(1-\mu)$; $\varepsilon_v = 3\varepsilon = 3\,p\,d/4tE(1-\mu)$.

Example 5

Thin cylinder $d = 1.2$ m, $t = 12$ mm, $p = 1.5$ N/mm^2, $L = 3$ m, $E = 2\times10^5$, $\mu = 0.3$.

$\sigma_c = 1.5 \times 1200/(2\times12) =$ **75 N/mm^2** ; $\sigma_l =$ **37.5 N/mm^2**

$\varepsilon_c = (75 - 0.3\times37.5)/2\times10^5 = 3.19\times10^{-4}$; $\varepsilon_l = (37.5 - 0.3\times75)/2\times10^5 = 7.5\times10^{-5}$

$\delta d = 3.19\times10^{-4} \times 1200 =$ **0.383 mm** ; $\delta L = 7.5\times10^{-5} \times 3000 =$ **0.225 mm**

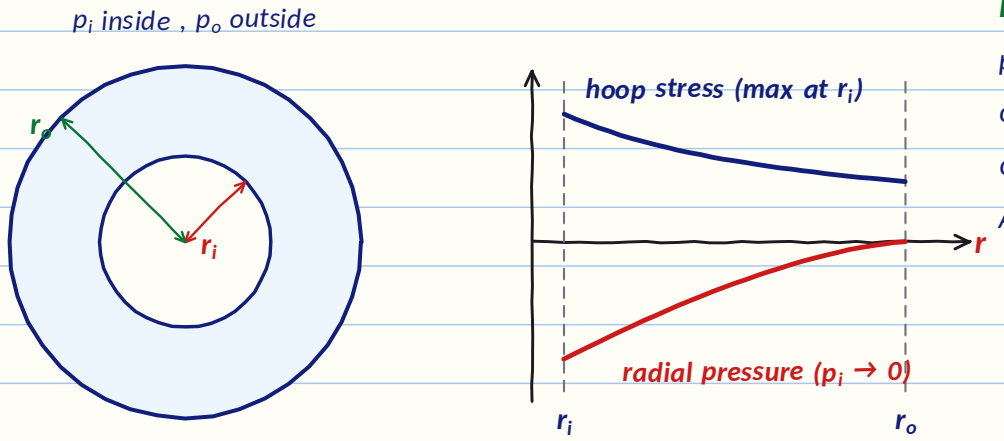
$\varepsilon_v = 2(3.19\times10^{-4}) + 7.5\times10^{-5} = 7.13\times10^{-4}$; $V = 3.393\times10^9$ mm^3 $\Rightarrow \delta V =$ **2.42×10^6 mm^3**

Example 6 (Thickness)

$d = 800$ mm, $p = 2.5$ N/mm^2, $\sigma_{perm} = 100$ N/mm^2 : $t = p\,d/2\sigma = 2.5\times800/200 =$ **10 mm**

5.6 Thick Cylinders – Lamé's Theory

Thick shell : $t > d/20$; hoop stress σ_h varies across the thickness and radial stress p_r acts.



Lamé :

$p_r = A/r^2 - B$

$\sigma_h = A/r^2 + B$

$\sigma_h + p_r = 2A/r^2$

A, B from B.C.

Assumptions

- Material homogeneous, isotropic, elastic ; plane sections remain plane (longitudinal strain constant) ; σ_l uniform.

Derivation of Lamé's equations

- Element between r and r + dr (angle dθ). Radial balance (p_r compressive +) : $p_r\, r\, d\theta = (p_r + dp_r)(r + dr)d\theta + \sigma_h\, dr\, d\theta$
- $\Rightarrow \sigma_h = -\left(p_r + r\, dp_r/dr \right)$ … (1)
- Constant longitudinal strain $\Rightarrow \sigma_h - p_r = 2B$ (constant) … (2). Put (2) in (1) : $r\, dp_r/dr = -2(p_r + B) \Rightarrow p_r + B = A/r^2$. Hence :

$$p_r = \frac{A}{r^2} - B \qquad \sigma_h = \frac{A}{r^2} + B$$

- Boundary conditions : $p_r = p_i$ at $r = r_i$; $p_r = p_o$ at $r = r_o$.

$$A = \frac{(p_i - p_o)\, r_i^2\, r_o^2}{r_o^2 - r_i^2} \qquad B = \frac{p_i\, r_i^2 - p_o\, r_o^2}{r_o^2 - r_i^2}$$

Only internal pressure ($p_o = 0$)

$$\sigma_h = \frac{p_i\, r_i^2}{r_o^2 - r_i^2}\left(1 + \frac{r_o^2}{r^2}\right) \qquad p_r = \frac{p_i\, r_i^2}{r_o^2 - r_i^2}\left(\frac{r_o^2}{r^2} - 1\right)$$

- Max hoop stress at the inner surface : $\sigma_{h,max} = p_i\,(r_o^2 + r_i^2)/(r_o^2 - r_i^2)$ ($> p_i$ always).
- At outer surface : $\sigma_h = 2\, p_i\, r_i^2/(r_o^2 - r_i^2)$; radial pressure falls from p_i to 0.
- Longitudinal stress (closed ends) $\sigma_l = p_i\, r_i^2/(r_o^2 - r_i^2) = B$.
- Design thickness for allowable σ_t : $r_o = r_i \sqrt{((\sigma_t + p)/(\sigma_t - p))}$.

Example 7

r_i = 100 mm, r_o = 150 mm, p_i = 80 N/mm^2, p_o = 0.

$B = 80 \times 10^4/(22500 - 10000) = 64$; $A = B\, r_o^2 = 1.44 \times 10^6$

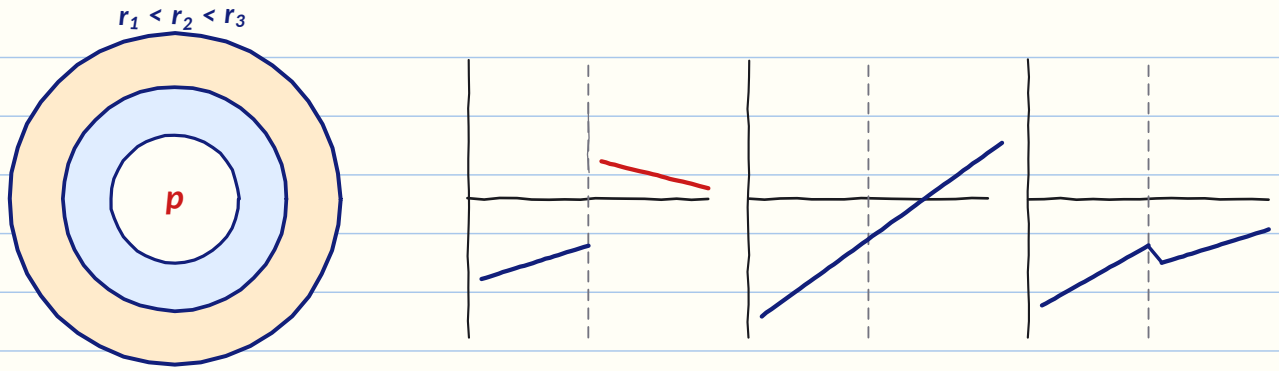
σ_h (inner) = $A/r_i^2 + B = 144 + 64 =$ **208 N/mm^2** ; σ_h (outer) = 64 + 64 = **128 N/mm^2**

p_r : 80 N/mm^2 (inner) → 0 (outer) ; σ_l = **64 N/mm^2**

Example 8 (Wall thickness)

Inner radius 80 mm, p = 40 N/mm^2, σ_t = 120 N/mm^2 : $r_o = 80\sqrt{(160/80)} = 113.1$ mm ⇒ **t = 33.1 mm**

5.7 Compound Cylinders



inner tube + outer jacket (shrunk on) hoop stress variation across the wall (schematic)

- Outer jacket (smaller bore) is heated and shrunk on the inner tube ⇒ inner tube is compressed, outer jacket is in hoop tension *before* loading.
- Resultant stress = shrinkage stress + stress due to working pressure (Lamé) ⇒ stress is far more uniform across the wall and material is better used.
- Junction pressure (shrinkage) p_s acts at r_2 ; radii r_1 (bore), r_2 (junction), r_3 (outside).
- Inner tube : σ_h at $r_2 = -\,p_s\,(r_2^2 + r_1^2)/(r_2^2 - r_1^2)$; at $r_1 = -\,2\,p_s\,r_2^2/(r_2^2 - r_1^2)$.
- Outer tube : σ_h at $r_2 = +\,p_s\,(r_3^2 + r_2^2)/(r_3^2 - r_2^2)$; at $r_3 = +\,2\,p_s\,r_2^2/(r_3^2 - r_2^2)$.
- Radial interference (same material) $\approx r_2\,(\sigma_{h,\text{outer-bore}} + |\sigma_{h,\text{inner-outside}}|)/E$.

Procedure

- 1. Draw Lamé distribution for the shrinkage pressure p_s (inner tube compressive, outer tube tensile).
- 2. Draw the Lamé distribution for the internal working pressure on the equivalent single cylinder.
- 3. Algebraically add the two ⇒ resultant hoop stress at each radius.
- 4. Choose p_s so that max resultant hoop stress is equal in both tubes (best design).

Example 9

$r_1 = 100$, $r_2 = 150$, $r_3 = 200$ mm ; $p_s = 20$ N/mm^2, $E = 2\times10^5$.

Outer tube : bore $20(40000 + 22500)/(17500) =$ **71.4 N/mm^2** (tension) ; outside $2\times20\times22500/17500 =$ **51.4 N/mm^2**

Inner tube : outside $-\,20(32500)/12500 =$ **– 52 N/mm^2** ; bore $-\,2\times20\times22500/12500 =$ **– 72 N/mm^2** (compression)

Interference $\approx 150\,(71.4 + 52)/2\times10^5 =$ **0.093 mm** on radius

Formula summary – Unit V

Item	Formula
Euler load	$\pi^2 EI / L_e^2$
Slenderness ratio	$\lambda = L_e / k$
Rankine	$\sigma_c A /(1 + a \lambda^2)$
Thin cylinder hoop	$\sigma_c = p\ d / 2t$
Thin cylinder long.	$\sigma_l = p\ d / 4t$
Lamé hoop stress	$\sigma_h = A/r^2 + B$
Lamé radial pressure	$p_r = A/r^2 - B$

Short questions

- Define slenderness ratio. A: effective length / least radius of gyration.
- Why does a column buckle about the weaker axis? A: least I gives least P_{cr}.
- Hoop stress vs longitudinal stress in thin cylinder? A: $\sigma_c = 2\ \sigma_l$.
- Where is the hoop stress maximum in a thick cylinder? A: at the inner surface.